

RESEARCH ARTICLE SUMMARY

AIR POLLUTION

Heavily polluted Tijuana River drives regional air quality crisis

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INTRODUCTION: The Tijuana River watershed is the epicenter of an environmental and public health crisis. For decades, millions of gallons per day of untreated sewage, industrial waste, and contaminated runoff have been released into the river, subsequently flowing through the Tijuana River Estuary and into the Pacific Ocean. These heavily contaminated waters recently caused nearly 1300 consecutive days of Southern California beach closures, negatively affecting both public health and the economy. Whereas health concerns have typically focused on direct exposure to contaminated waters, this study addresses the often-ignored impact of polluted waters on air quality. Pollutants are transferred from water to air—a process that is enhanced by turbulent flows—where they can be dispersed over tens of kilometers. As water pollution increases globally, understanding the atmospheric consequences of contaminated water bodies is critical to safeguarding public health.

RATIONALE: In summer 2024, even in the absence of rain, wastewater flows surged to millions of gallons per day, enhancing water-to-air transfer of hydrogen sulfide (H_2S)—a toxic gas and key sewage tracer—and other hazardous gases. Despite community concerns, the pollutants causing malodors and adverse health effects had never been identified or quantified. In this study, we combined advanced measurements and trajectory modeling to demonstrate that poor water quality can substantially affect air quality, highlighting inhalation as a major exposure pathway while validating long-dismissed community voices.

RESULTS: Neighborhood-based nighttime H_2S levels near a riverine hotspot located at the Saturn Boulevard river crossing peaked at 4500 parts per billion (ppb)—with hourly averages orders of magnitude above typical urban levels (<1 ppb) and California's 1-hour ambient air quality standard (30 ppb). A sudden wastewater diversion abruptly reduced peak nighttime flows from between 40 and 80 million gallons

per day (MGD) to <5 MGD. H_2S levels and odor reports were strongly correlated (correlation coefficient $r = 0.92$), both sharply decreasing (~95%) after the wastewater diversion. The strong correlation supports that the turbulent riverine hotspot was the dominant source of malodors and H_2S . Air samples collected before and after the diversion also identified numerous compounds with known toxicity that decreased in abundance after the wastewater diversion, underscoring that exposure is not limited to a single toxic gas. An air trajectory model was able to reproduce the observed H_2S trends only when meteorology and river flows were represented, further supporting the riverine hotspot as the primary source. The model demonstrated the potential for widespread dispersion of airborne pollutants well beyond the edge of the contaminated Tijuana River.

CONCLUSION: Communities along the Tijuana River Valley have endured decades of environmental pollution, with a public health crisis now clearly linked to airborne exposure of wastewater-derived pollutants. Such prolonged exposure without adequate protection or timely intervention highlights an environmental injustice that demands action. Immediate actions—such as diverting wastewater away from the Tijuana River and implementing localized air monitoring—can sharply reduce toxic exposures, but long-term solutions require upstream treatment infrastructure, strengthened regulatory oversight, and binational collaboration to reduce pollutant discharges at their source. As the number of heavily polluted water bodies grows with ever-increasing inputs, ignoring water-to-air contaminant emissions in air quality management allows hidden risks to persist. Recognizing and addressing this airborne exposure pathway is critical to protecting vulnerable populations and advancing global health equity. □

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Turbulence in polluted waters transfers contaminants to the air. Culverts at the Saturn Boulevard river crossing generate high turbulence, enhancing water-to-air transfer of toxic wastewater pollutants.



AIR POLLUTION

Heavily polluted Tijuana River drives regional air quality crisis

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Industrial chemicals and untreated sewage have polluted the Tijuana River for decades, recently causing >1300 consecutive days of California beach closures. In summer 2024, wastewater flows surged to millions of gallons per day despite no rain, enhancing water-to-air transfer of hydrogen sulfide (H₂S) and other toxic gases at a turbulent hotspot. High wastewater flows and low winds led to nighttime H₂S peaks, reaching 4500 parts per billion (ppb)—exceeding typical urban levels of <1 ppb. H₂S levels and community malodor reports were strongly correlated (correlation coefficient $r = 0.92$), validating long-dismissed community voices and highlighting an environmental injustice. This study demonstrates that poor water quality can substantially affect air quality—although rarely included in air quality models and health assessments—with far-reaching implications as polluted waterways increase globally.

The 4532-km² Tijuana River Valley, spanning the western US-Mexico border, has been the epicenter of a worsening environmental and public health crisis over many decades. The heavily polluted Tijuana River, containing untreated sewage, industrial waste, harmful chemicals, and contaminated stormwater, crosses the border and flows through the Tijuana River Estuary before emptying into the Pacific Ocean at the southern edge of San Diego County (1, 2). The Tijuana watershed has been classified as an impaired water body under the US Clean Water Act, with >1300 days of recent beach closures and considerable impacts on public health, the environment, and the region's economy (3). In 2025, the Tijuana River ranked second on the “America's Most Endangered Rivers” list (4). During the rainy season, November through March, river flows can reach billions of gallons per day. Until recently, dry-season flows (June through September) have been negligible except after rare summer rain events. The ongoing water pollution crisis in the Tijuana River watershed has been exacerbated by the winter arrival of record high numbers of intense atmospheric rivers in 2023 and 2024, damaging already aging and inadequate wastewater infrastructure on both sides of the border (5, 6). Coastal discharge of treated and untreated sewage coming from Punta Bandera—a major wastewater outfall located in Mexico, 10 km south of the border—also flows northward during the southern swell, adding to the polluted coastline from Imperial Beach north to other coastal San Diego communities (7–10). Additionally, millions of gallons of untreated industrial waste containing heavy metals and harmful chemicals are released into the Tijuana River from Maquiladoras, duty- and tariff-free manufacturing facilities in Mexico just south of the US-Mexico border (11, 12).

Health concerns regarding pollution in the Tijuana River watershed have focused primarily on direct exposure to contaminated water—either by drinking or by entering the ocean for recreational activities (13–15). However, many processes can transfer water pollutants to the

air, including bursting bubbles in waterfalls, turbulence in rivers, aeration in wastewater treatment plants, toilet flushing, and breaking waves in coastal surf zones, which form aerosols in a process known as aerosolization (16–19). Studies have shown that water pollutants, bacteria, viruses, pathogens, and fungal spores can become enriched in the aerosolization process (20–27). In addition to aerosolization, gaseous pollutants can partition from water to air (28, 29). Airborne pollutants can then be dispersed by winds over distances of tens of kilometers, increasing the number of people exposed to these pollutants beyond just those in direct contact with the water. Furthermore, humans inhale ~11,000 liters of air daily—far more than the 2 liters of often-filtered water they drink—making inhalation potentially the primary exposure pathway for many water-derived pollutants (30, 31). The impact of heavily contaminated waterways on air quality and human health has long been underestimated and largely ignored, with increasing consequences as water pollution, anthropogenic runoff, and local flooding are increasing as a result of population growth, aging infrastructure, and climate change (32).

This study offers a real-world example of how pollutant exposure can extend well beyond the banks of a contaminated river or waterway. The health impacts of pollutants and pathogens transferred from heavily contaminated waters into the atmosphere remain a substantial and largely unaddressed gap in our understanding (33, 34). Given that more than half of the global population resides near rivers, lakes, or coastlines (35), strategic measurements and modeling are needed to monitor and map the ways that pollution moves from water bodies to surrounding air and, ultimately, to local communities and beyond. More broadly, the impact of water pollution on air quality, including the impacts of emerging air contaminants of concern, represents a critical yet underexplored public health issue with major implications for all of humanity (36).

Identifying the Tijuana River hydrogen sulfide and malodor hotspot

In a previous winter season field study at Imperial Beach, we used genomic sequencing to show that up to 76% of airborne bacteria could be directly traced back to the contaminated Tijuana River (37). In an additional winter season study in 2020, illicit drugs, drug metabolites, and chemicals from tires and personal care products were detected in the South Bay, San Diego, region in both water and aerosol samples (38). The field study discussed here has broadened the focus to include gas and aerosol measurements in the same region during the summer, a period when Tijuana River flows were expected to be minimal.

Air quality measurements commenced at a coastal site near the US-Mexico border, Border Field State Park (Fig. 1, sampling location 1), which in the summer can be affected by ocean water polluted by outfall from Punta Bandera (9). However, input from residents in communities several kilometers from the coast—including reports of malodors and health symptoms—helped us identify a turbulent and foamy site along the Tijuana River that was visible in satellite imagery (Fig. 1 and figs. S1 and S2). As this site was the only visibly turbulent portion of the Tijuana River on the US side of the border, and on the basis of findings in wastewater management studies, we hypothesized that the turbulence at this site was leading to enhanced emissions of hydrogen sulfide (H₂S) and other gases, aerosols, and associated malodors (39–43), thereby creating a riverine hotspot. Consequently, on 1 September 2024, we moved our mobile air quality laboratory ~5 km inland to the community of Nestor in the city of San Diego, the closest neighborhood to the riverine hotspot and near a high density of K–12 schools (fig. S3). We focus here on the H₂S measurements taken at Border Field (Fig. 1, sampling location 1) and Nestor (Fig. 1, sampling location 2) and volatile organic compounds (VOCs) collected on sorbent tubes at the Saturn Boulevard hotspot (Fig. 1, sampling location 3), the Tijuana River Estuary (Fig. 1, sampling location 4), and a coastal site at Imperial Beach (Fig. 1, sampling location 5).

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H₂S: A toxic sewage gas and airborne tracer of water pollution

For decades, communities near the Tijuana River Valley have reported malodors and health symptoms (44). Despite overwhelming community concerns, comprehensive air quality measurements had never been made to identify the specific pollutants causing the malodors and the adverse health effects. Known for its distinctive “rotten egg” smell and low odor threshold (45), H₂S is the most common gaseous tracer of sewage. In this study, we used H₂S to link malodors and other airborne contaminants with sewage-laden waters. Produced by the anaerobic decomposition of organic matter, H₂S is toxic even at relatively low levels (46). The Centers for Disease Control and Prevention (CDC) Agency for Toxic Substances and Disease Registry (ATSDR) reports that urban ambient H₂S levels are typically <1 part per billion (ppb) (47, 48). The California Air Resources Board (CARB) set a 1-hour average California Ambient Air Quality Standard (CAAQS) of 30 ppb for the purpose of odor control. The South Coast Air Quality Management District (South Coast AQMD) uses this same 30-ppb level for acute exposure. The California Office of Health Hazard Assessment (OEHHA) and South Coast AQMD use a chronic exposure limit of 7.3 ppb, particularly for vulnerable populations (e.g., pregnant women, children, older adults, asthmatics, the immunocompromised).

Figure 2 shows 1-hour average H₂S levels at the Border Field coastal site (top row) as well as the Nestor neighborhood site (bottom row). Measured levels are shown relative to the 30 ppb 1-hour average CAAQS for H₂S. During the 24–25 August sampling period at Border Field, H₂S levels remained below the CAAQS (Fig. 2, A to D). From 7 to 8 September at the Nestor site, H₂S levels routinely exceeded the CAAQS. The polar plots clearly demonstrate that the proximity of Nestor to the riverine hotspot led to much higher H₂S levels than at Border Field, particularly during low-wind, stagnant-air nighttime conditions (Fig. 2, E and F).

Unprecedented dry 2024 summer season flows

In the past two rainy seasons, unusually high numbers of winter storms with intense rainfall resulted in record transboundary annual Tijuana River flows of 44 and 36 billion gallons in 2023 and 2024, respectively (Fig. 3A and fig. S4). The two corresponding dry seasons also had higher annual flows than previous years (Fig. 3, B to F). The impact of Hurricane Hilary in the summer of 2023 resulted in higher-than-normal August flows (Fig. 3E). Whereas high rainy season flows reflect contribution and major dilution from rainfall, dry-season flows are typically low, with minimal contribution and dilution from rainfall (49). As shown in Fig. 3F, unprecedented high river flows were recorded throughout most of the 2024 dry season, ranging between 40 and 80 million gallons per day. The release of millions of gallons of wastewater and untreated sewage into the Tijuana River stemmed largely from an infrastructure failure at the Punto de Bombeo de Control Integral Lomas de Agua (PBCILA). Although flow volume data alone do not pinpoint the exact date of the PBCILA pump failure, a sharp increase in community odor complaints indicates that the riverine hotspot—characterized by high dry-season flows (low dilution, untreated sewage) and elevated turbulence—was active by July 2024. On 10 September 2024, the PBCILA facility began diverting wastewater to the San Antonio de los Buenos wastewater treatment plant, where it was then released into the Pacific Ocean at Punta Bandera.



Fig. 1. Map of air sampling sites. The red pins indicate H₂S sampling locations at (1) the coastal sampling site at Border Field State Park and (2) the inland sampling site in Nestor community (city of San Diego) near the riverine hotspot. Blue pins indicate VOC sampling locations at (3) the river crossing hotspot at Saturn Boulevard, (4) the Tijuana River Estuary, and (5) a coastal site in Imperial Beach. The red box shows the river crossing hotspot at Saturn Boulevard, identified with community input as the major source of malodors. The inset highlights the region where turbulent flow produces extensive surface foam (photo taken on 5 September 2024).

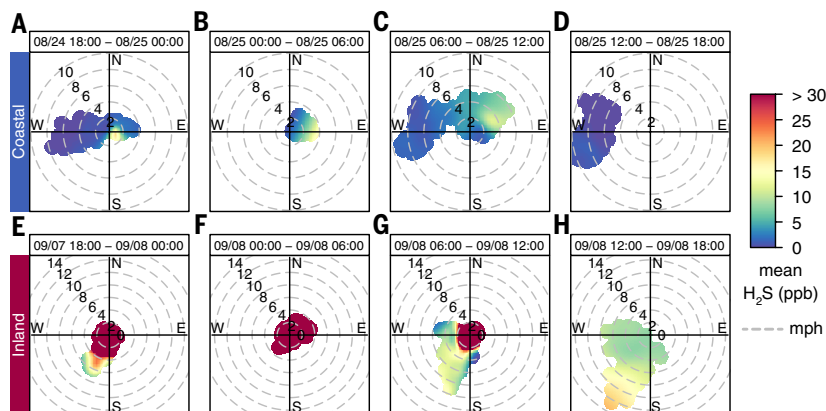


Fig. 2. Polar heatmaps of H₂S levels at coastal and inland sampling sites segmented into four 6-hour periods. (A to D) H₂S polar heatmaps at the Border Field State Park coastal site, spanning 24 to 25 August 2024. (E to H) H₂S polar heatmaps at the inland site of Nestor, spanning 7 to 8 September 2024. Color bar denotes 1-hour average H₂S levels with a maximum scale value of 30 ppb, corresponding to the CAAQS. Windspeeds shown in miles per hour (mph).

This diversion of wastewater effectively reduced transboundary flows and associated turbulence at the riverine hotspot, with daytime flows dropping to nearly zero and nighttime flows to <5 million gallons per day. High-resolution 15-min flow data confirm that, after the abrupt diversion on 10 September, wastewater was discharged to the river only between 6 p.m. and 4 a.m. for the remainder of the 2024 dry season (Fig. 3F, inset, and fig. S5).

Record high dry-season flows drive toxic H₂S levels and community malodor reports

Measurements were conducted in the Nestor community in the city of San Diego to quantify the levels of airborne H₂S in this residential community from the dry-season Tijuana River turbulent hotspot. See

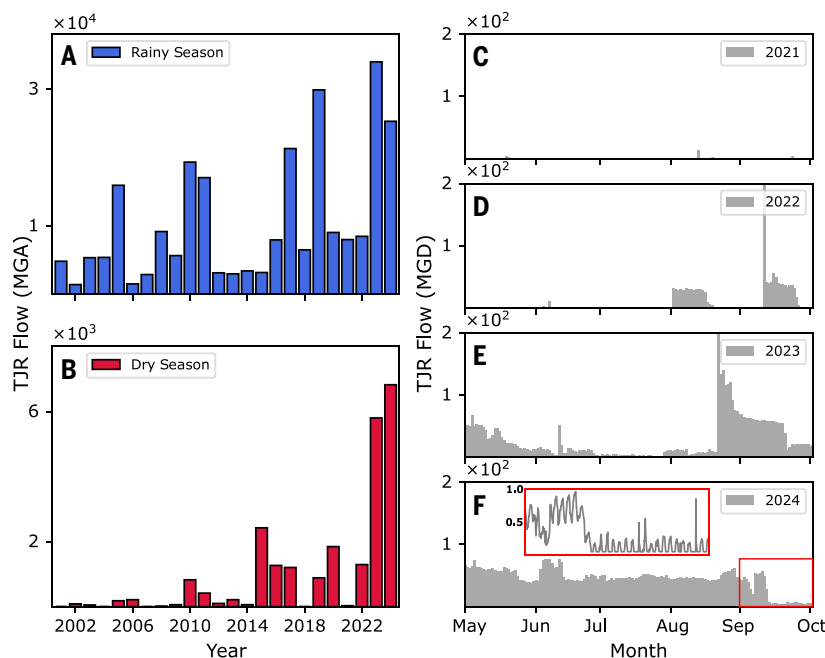


Fig. 3. Historical annual transboundary Tijuana River flow. (A) During the San Diego rainy season (November through March) between January 2001 and October 2024 in millions of gallons annually (MGA). (B) During the San Diego dry season (May through September) between January 2001 and October 2024. (C to F) Annual dry-season Tijuana River flow between 2021 and 2024. Inset in (F) shows normalized Tijuana River flow (15-min averaged) between 1 and 30 September 2024. Data were downloaded from the International Boundary and Water Commission (IBWC) database (78).

fig. S6 for a summary of the full timeline of events related to this study. Time series of H_2S levels measured at Nestor from 1 to 22 September 2024 are shown in Fig. 4. During the period with the highest river flows (5 to 9 September), 1-min averaged H_2S levels surged at night to 4500 times the typical urban ambient levels (Fig. 4A). After the 10 September wastewater diversion, H_2S levels decreased nearly 95%. However, H_2S levels still peaked each night, coinciding with controlled nighttime releases that increased river flow and turbulence at the hotspot, while wind speeds and temperatures were at their lowest (figs. S7 to S9). Meteorological conditions led to reduced dispersion and atmospheric dilution of H_2S , trapping the gas (which is heavier than air) near the ground in nearby communities between ~6 p.m. each night and ~8 a.m. the next morning (figs. S10 and S11).

Residents living in south San Diego and Imperial Beach communities made increasing and ultimately record-high numbers of daily malodor reports (Fig. 4B) to the San Diego County Air Pollution Control District (SDAPCD). These malodor reports were typically made during the nighttime or early morning hours. The sudden wastewater diversion and corresponding decrease in H_2S levels on 10 September were directly reflected in malodor reports, which decreased from 233 daily reports on 9 September to fewer than 10 by 12 September. Linear regression analysis reveals a strong positive correlation ($r = 0.92$) between H_2S levels and community malodor reports (fig. S12). The strong temporal correlation between the polluted Tijuana River flow and ambient H_2S levels in Nestor supports the hypothesis that the turbulent riverine hotspot was the dominant source of malodors and H_2S emissions.

Framing the H_2S measurements in a regulatory context, 1-hour average H_2S levels reached 2100 ppb (Fig. 4B), nearly 70 times the 1-hour CAAQS of 30 ppb. Figure 4C illustrates the number of hours in a 24-hour period that residents in Nestor were exposed to H_2S levels above the CAAQS. Before the PBCILA facility diversion, residents experienced

between 5 and 14 hours of exposure above the CAAQS, largely at night, with 1-hour average H_2S levels ranging from 30 to 2100 ppb. After the wastewater was diverted, despite H_2S levels decreasing significantly, residents still experienced up to 5 hours of exposure above the CAAQS on most nights (Fig. 4, B, inset, and C). The OEHHA/South Coast AQMD chronic exposure limit of 7.3 ppb is also included in Fig. 4B for reference. Although this study enables direct comparison with acute exposure standards, evaluating chronic exposure will require forthcoming long-term measurement data to determine whether these levels reflect typical air quality conditions over months to years.

Dispersion of air pollutants broadens Tijuana River impact to regional scale

The high temporal resolution H_2S measurements in this study capture the impact of the high dry-season contaminated flows on Nestor, the local city of San Diego community closest to the turbulent riverine hotspot. To evaluate our emissions assumptions, and to investigate the extent to which pollution from the Tijuana River is affecting air quality in neighboring communities and beyond, the HYSPLIT (50) trajectory and dispersion modeling platform was used. The 1-hour average Tijuana River flow rates were used as a semiquantitative proxy for relative H_2S emissions, capturing the effect of increased flow on source strength as well as the critical role of turbulence at the riverine hotspot.

Dispersion modeling allows further study of the factors affecting the spatial distribution of airborne pollutants emanating from the riverine hotspot, including source strength, source location, and meteorology.

Figure 5 illustrates the modeled regional impact of H_2S and likely other toxic gas emissions from the Tijuana River crossing at Saturn Boulevard. Relative levels of potentially degraded air masses were binned into three time periods with distinct flow and meteorological conditions: high river flow during moderate-temperature conditions from 1 to 5 September (period 1); high river flow during atmospherically stagnant conditions from 6 September through 12:00 p.m. on 10 September (period 2); and reduced river flow with westerly winds over the remainder of the study from 12:01 p.m. on 10 September through 22 September (period 3). The cutoff between periods 2 and 3 was set to noon on 10 September, because the wastewater diversion occurred that morning.

As shown in Fig. 5, high Tijuana River flow periods at the beginning of September resulted in higher airborne pollutant levels near the riverine hotspot source region (shown as a black point in all panels). Compared with period 1, period 2 nighttime boundary layer winds were weaker, resulting in the highest H_2S buildup in and around Nestor and surrounding communities (Fig. 5, A and B). Higher river flows starting in the evening hours reproduced the nightly increases of H_2S in Nestor, showing earlier and higher levels in period 2 (Fig. 5, D and E). Reduced flows occurring after the wastewater diversion led to reduction in turbulence and H_2S emissions during period 3. Along with more consistent westerly winds, reduced river flow contributed to a greatly reduced density of weighted trajectories over the study region, with nighttime peaks only a fraction of those occurring during high river flow conditions (Fig. 5, C and F). Note that weighting summed trajectories by the 1-hour average river flows was necessary to reproduce the measured trends at Nestor across the three time periods (i.e., meteorology alone without flow-based weighting did not reproduce the trends), further confirming the Saturn Boulevard riverine hotspot as the primary source region driving local emissions of H_2S and other airborne pollutants.

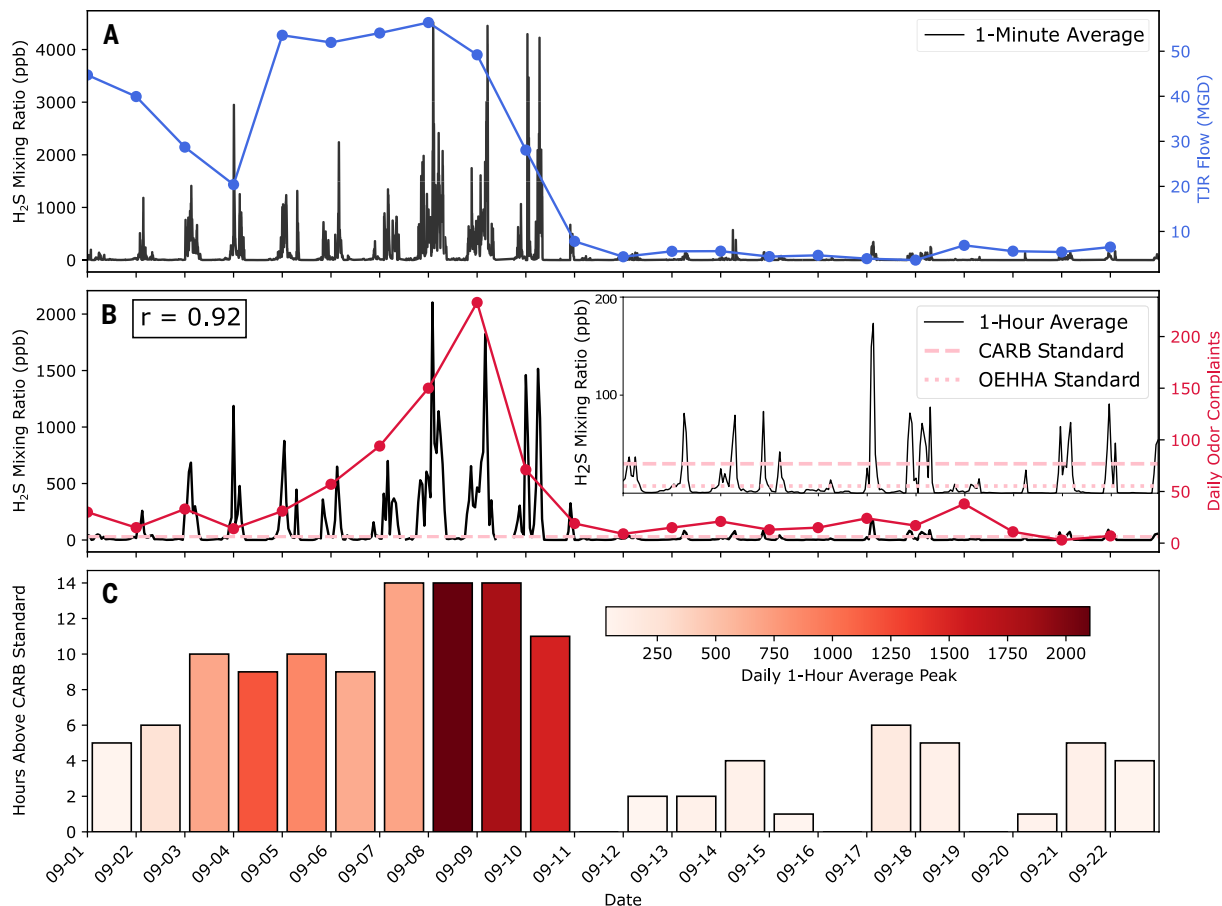


Fig. 4. Time series for H₂S, malodor reports, and Tijuana River flow with comparison to air quality standards. (A) H₂S levels (1-min average) and daily Tijuana River flow in millions of gallons per day (MGD). (B) H₂S levels (1-hour average) and daily malodor reports received by the SDAPCD. Correlation ($r = 0.92$). Dashed light-red line shows the CAAQS of 30 ppb. Dotted light-red line shows OEHHA standard for chronic exposure of 7.3 ppb. (C) Bar plot of the number of hours per day ambient H₂S levels exceeded the 1-hour average CAAQS. The color bar indicates daily peak H₂S levels in parts per billion (1-hour average).

H₂S indicates contaminated waters as a source of additional airborne pollutants

The wastewater flowing through the Tijuana River, signaled by the malodorous H₂S, produced a multitude of airborne trace gases, some of which are known air toxics. Abundances of VOCs sampled using sorbent tubes were compared in paired samples, collected before and after the 10 September wastewater diversion, at each of three locations: the Saturn Boulevard river crossing, the Tijuana River Estuary, and the Imperial Beach coast (Fig. 1). At each location, the paired samples were collected at approximately the same time of day under the same meteorological conditions (table S1) to better isolate the influences of different source contributions. Figure 6 highlights select gaseous oxygenated hydrocarbons (CHO; see table S2 for compound identification) that showed a decrease in percentage abundance after diversion at the riverine hotspot (Fig. 6A), the Tijuana River Estuary (Fig. 6B), and a coastal site in Imperial Beach (Fig. 6C). Analogous figures are provided in the supplementary materials for hydrocarbons (CH; fig. S13 and table S3) and heteroatom-containing hydrocarbons (CHX, where X denotes sulfur, nitrogen, bromine, or chlorine; fig. S14 and table S4). Diurnal profiles of the selected compounds sampled at the Saturn Boulevard site (fig. S15) largely show peak abundance in the night and early morning samples, coinciding with observed increases in emissions due to the nightly releases and increases in transboundary flow and associated turbulence at the riverine hotspot, coupled with lower wind speeds and reduced atmospheric mixing.

Figure 6 uses a color-coded scale based on level-two 1-hour Acute Exposure Guideline Limits (AEGLs, or their equivalents), which indicate concentrations associated with irreversible or other serious, long-lasting adverse health effects (57). These colors reflect the presence of a compound with an established level-two AEGL (AEGL-2) but do not represent measured levels (AEGL-1 and AEGL-3 values are given in tables S2 to S4). Quantification studies are underway to assess whether compounds were present above these acute thresholds. White bars denote compounds with no established acute thresholds; however, such chemicals may still pose health risks, particularly when part of a complex mixture. The presence of compounds with established AEGLs highlights that hazardous airborne pollutants extend beyond H₂S. Because AEGLs address only short-term exposures, comprehensive assessments are needed to evaluate both acute and chronic health risks from the distinctive and complex chemical mixtures emitted from the Tijuana River.

The observed decreases in compound abundance across sites reflect differences in meteorology, photochemistry, and, particularly at the coastal site, greater influence from nonriver sources. Samples taken at the Tijuana River Estuary were collected in the early morning, with minimal influence of photochemistry and/or dispersion, whereas samples taken from Saturn Boulevard were collected in the afternoon, when both processes were more active. The large reduction in compound abundance at the Tijuana River Estuary after the wastewater diversion further supports that the contaminated Tijuana River flow

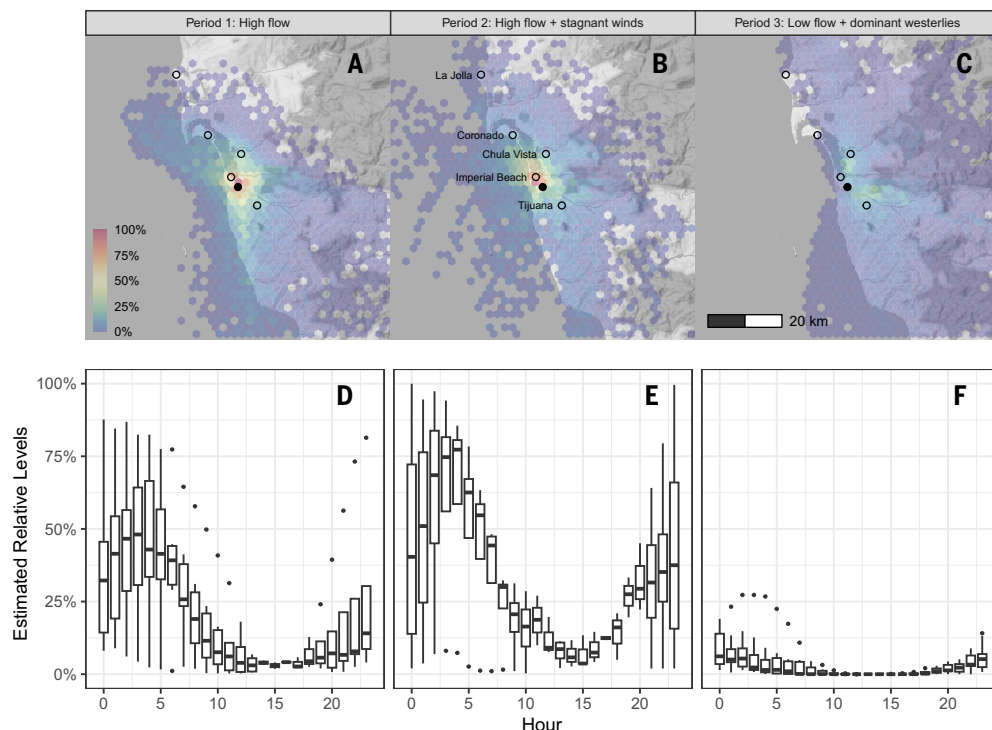


Fig. 5. Spatial and diurnal patterns for flow-weighted trajectory model end points starting at the Saturn Boulevard Tijuana River crossing H_2S hotspot. Relative dispersion and spatial intensities across Tijuana River Valley and surrounding area are shown for (A) period 1 (1 to 5 September), (B) period 2 (6 to 10 September), and (C) period 3 (1 to 23 September). The cutoff between periods 2 and 3 was set at 12:00 p.m. on 10 September 2024. (D to F) Hourly sums across San Diego and Imperial Beach communities are shown for periods 1, 2, and 3, respectively. All values are scaled to their respective maxima, providing an estimate of relative concentrations based on local meteorology and assumed emission behavior.

was the main contributor to the selected CHO, CH, and CHX gaseous species at this site. In contrast, the more variable reduction at the coastal site reflects a wider range of nonriver sources contributing to the selected gas-phase compounds. The differences in source contributions are also reflected in the relative abundance of each compound before diversion (fig. S16), where the coastal site shows the lowest fractional contribution for nearly all compounds, except for a few CHX species likely with dominant ocean sources. Given the combined influence of source variability, photochemistry, and meteorological conditions, coupled field observations and atmospheric modeling will be needed to provide quantitative assessments of these gases on air quality and community exposure.

Polluted water bodies as emission sources for air quality modeling

This study demonstrates the major impact a turbulent portion of a contaminated river can have on regional air quality—a finding with global implications, as many countries grapple with waterways contaminated by increasing levels of untreated sewage, chemical runoff, and industrial waste. The severity of water pollution in the Tijuana River was visibly marked by foam accumulating on the river surface, peaking each morning and often persisting throughout the day (figs. S1 and S2), starting at the Saturn Boulevard hotspot and extending through the Tijuana River Estuary. A global example of similar persistent foam layers has been reported in the Yamuna River in New Delhi, which has been attributed to a similar combination of industrial pollutants and untreated wastewater (52, 53). These surface foams have been shown to contain untreated waste, industrial chemicals, PFAS, metals, and surfactants that accumulate on the water's surface (52–54). Similar foaming and enhanced H_2S emissions have been observed in

gravity sewer systems, where the vertical drops create turbulence that drives efficient water-to-air transfer of H_2S from highly concentrated wastewater (39–43).

Air trajectory modeling supports the recognition of polluted transboundary river flow rates—which serve as a proxy for source strength of water-derived pollutants—alongside meteorological variability as key drivers behind observed diurnal patterns and levels of H_2S . It is important to reiterate that H_2S is only one of potentially thousands of other pollutants emitted from the Tijuana River, some of which are reported here, that can affect regional health across San Diego County and beyond (55). Both measurements and models demonstrate that changes in wind speed and direction, on time-scales of hours to days, influenced the variability of H_2S and other co-emitted pollutants. Therefore, routine air quality measurements that capture these spatial and temporal variations, especially near turbulent zones with waterfalls, sewage drop-offs, and other structures producing hydraulic jumps and rapids, are needed to serve as inputs for air quality models, assess neighborhood- and regional-scale air quality, and address public health concerns. Such measurements are critical for advancing current esti-

mates of the public health consequences of water pollution, which do not typically include potential inhalation exposure pathways and degradation of air quality (56).

This study underscores the pressing need to enhance our understanding of how pollutants move between environmental reservoirs, in this case polluted water and air. Current air quality models do not include gaseous emissions from polluted rivers, estuaries, or coastal systems—areas increasingly affected by anthropogenic pollution. Incorporating emissions from contaminated water bodies will be essential for accurate predictions of health impacts of environmental pollution and developing effective mitigation strategies. As human population growth continues to drive pollution to levels that strain and exceed Earth's natural capacity for processing, understanding these transfer mechanisms is essential for addressing the regional and global impacts of pollution.

Addressing environmental justice and protecting public health

Residents living near the Tijuana River Valley have endured the impacts of environmental water pollution since the 1930s, with escalating impacts in recent years (44). The H_2S levels recorded in this study corroborate the long-standing concerns expressed by these residents. Despite experiencing chronic levels of H_2S exposure comparable with other communities, and potentially for considerably longer duration, their issues have received less attention and slower acknowledgment. A parallel can be drawn with the air quality crisis in Carson, California, in 2021 (57–59), where residents were subjected to malodors caused by high H_2S levels from a chemical spill in the Dominguez channel. In that emergency, the South Coast AQMD responded rapidly by conducting air monitoring, investigating sources, and issuing notices of violation, while the county provided air filtration devices and local

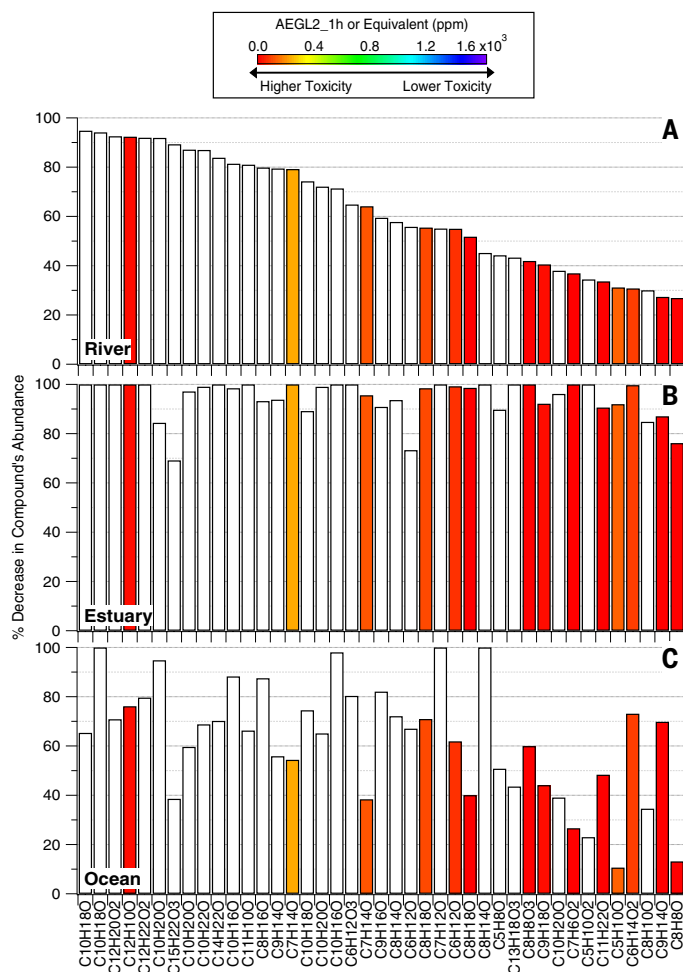


Fig. 6. Percent decrease in compound abundance after wastewater diversion.

Gaseous oxygenated hydrocarbons (CHO species) showing decreases after diversion of the wastewater in air samples collected at the riverine hotspot at Saturn Boulevard (A), the Tijuana River Estuary (B), and a coastal site in Imperial Beach (C). Bars are color coded on the basis of known compound-specific level-two 1-hour Acute Exposure Guideline Limits (AEGLs, or AEGL equivalents); white bars indicate compounds with no known AEGL or equivalent.

authorities worked to identify and eliminate the H_2S source. The South Coast AQMD has developed a network of real-time measurements of multiple toxic gases, including H_2S , across Los Angeles County, a publicly accessible display, and rapid-response communication to warn communities when high levels are reached to reduce community exposure (60). Although San Diego County has historically not undertaken the same level of air monitoring, source investigation, or community protection, the SDAPCD District has recently added three H_2S monitors to strengthen local surveillance efforts. However, the binational nature of the Tijuana River pollution crisis continues to pose major challenges for coordinated and sustained response.

Exposure to H_2S and other air pollutants released from untreated wastewater can help account for many of the long-standing health symptoms reported in South Bay, San Diego, and Imperial Beach, including respiratory and gastrointestinal issues, headaches, fatigue, skin infections, nausea, and anxiety (61). Wastewater treatment workers and residents near such facilities have experienced similar health effects (62–64). Additionally, studies on chronic H_2S exposure in near-source communities have shown an increased risk of noninfectious respiratory symptoms and disease, respiratory mortality, and chronic

bronchitis as well as ocular, nasal, and neurological effects (65–68). Studies focusing solely on H_2S do not fully represent the health impacts associated with the complex, transient mixtures of airborne pollutants that communities near polluted waterways are exposed to—especially young children, the elderly, and those with preexisting health conditions (56, 69). Given that ethical concerns have limited human trials on H_2S exposure (65), conducting controlled studies on human exposure to these harmful airborne mixtures would not be allowed. Nonetheless, this has essentially been the reality for the people living in these polluted environments for years. In October 2024, the CDC completed a Community Assessment for Public Health Emergency Response (CASPER) survey to investigate the health concerns and impacts of the Tijuana sewage crisis on nearby communities (61). This study completed 189 in-person household surveys over a 3-day period. Per the survey, 79.7% of households report that the sewage crisis has negatively affected their quality of life, 64.1% of households report worsening health conditions, 58% of households report that overall stress levels have increased because of the sewage crisis, and 70% of households report their symptoms caused by the sewage crisis improved when they left the area. Additionally, 89.3% of households believe that the crisis is worsening.

The demographics of Imperial Beach and the city of San Diego communities of Nestor and San Ysidro (Fig. 1) align with the definition of environmental justice communities (70). These areas have high Hispanic/Latino populations, ranging from 53 to 93%, and, a significant portion of residents live below the poverty line (71, 72). Besides the social stressors that the community endures, environmental pollution leads to cumulative impacts (73) on the local economy, health, and overall quality of life. Such prolonged exposure without adequate protection or timely intervention highlights a clear environmental injustice that would likely not be tolerated in more affluent or less marginalized communities.

Although addressing the scale of environmental pollution and associated health effects in the Tijuana River Valley will take some time, there are some immediate steps to be taken. Given that H_2S levels and other toxic gases peak at night, providing effective indoor air filtration represents a practical short-term solution. For the filters to be effective, they must include activated carbon with potassium permanganate (74) to reduce H_2S and other toxic gases, along with particulate filters to reduce exposure to aerosolized pollutants including viruses, bacteria, and fungal spores. In addition to filtration, implementing a community air monitoring program—similar to that established by the South Coast AQMD—is essential for protecting public health in the communities affected by atmospheric pollutants emanating from the Tijuana River. Such measures are fundamental to safeguarding community health by providing real-time data on air quality and enabling timely responses to pollution events (60, 75). Full assessment of the health impacts will require analyzing medical records, existing survey data, and continuous measurements of a range of airborne pollutants, and ultimately a large-scale epidemiological study in the communities affected by the heavily polluted Tijuana River will be necessary.

Making air quality measurements and indoor filtration accessible—paired with community education—empowers residents to protect themselves. However, long-term relief will only come from stopping pollution at the source. This study makes it clear that water pollution does not end at the river's edge. The toxic emissions we measured in the air are a direct result of untreated sewage and industrial waste being dumped into the Tijuana River. While long-term infrastructure upgrades and binational policy reforms are critical, they are projected to take more than 5 years to implement. In the meantime, an immediate and low-cost mitigation step would be to eliminate the turbulent drop at the Saturn Boulevard site, where the largest source of H_2S and other toxic gas emissions was identified. Without urgent interventions, communities on both sides of the border will continue to bear the health burden of this preventable crisis. Sustained monitoring,

coordinated cross-border efforts, and leadership from federal, state, and local authorities are crucial to finally provide the protection and justice long denied to communities affected by this ongoing environmental and public health crisis.

Materials and methods

H₂S analysis

Ambient H₂S levels were measured using a Teledyne T101 analyzer in a mobile laboratory stationed from 24 to 25 August 2024 at Border Field State Park and 1 to 22 September 2024 in the community of Nestor, 950 yards from a hotspot identified at the Tijuana River crossing at Saturn Boulevard. The T101 H₂S analyzer was operated in “H₂S-only” mode to provide maximum temporal resolution with a 1-min sampling rate. The T101 analyzer was operated in the 0 to 5 parts per million (ppm) full-scale mode. Before field deployment, the T101 was calibrated at the Teledyne API Factory in San Diego, California. During the campaign, cross-validations were performed using a Thermo 450iQ H₂S analyzer (fig. S17). A post-field study calibration was performed in the laboratory using a certified 25 ppm H₂S standard, purchased from Airgas (Airgas part number: X02NI99CA160000) that confirmed data accuracy and instrument stability throughout the study (fig. S18).

Meteorological data

Meteorological data, including temperature, wind speed, and wind direction were collected using colocated MODULAIR air quality monitors equipped with sonic anemometers from QuantaQ Inc. The sonic anemometers provide high-resolution wind measurements necessary for understanding the localized atmospheric dynamics that contribute to the observed H₂S levels. OpenAir plotting package was used to generate pollution heatmaps using hourly wind speed, wind direction, and H₂S data (Fig. 2).

River flow data

River flow data were obtained from the transboundary flow gauge located near the US-Mexico border, ~3 miles upstream from the Saturn Boulevard crossing, resulting in an estimated lag time of arrival of 3 to 4 hours. High-resolution average river flow data in millions of gallons per 15 min were downloaded from the IBWC database (Fig. 3F, inset). The daily averaged river flow data are shown in Fig. 4A. Annual total river flow from 1 January 2000 to October 2024 are shown in fig. S4. Accumulated river flow from 2021 to 2024 data are shown in fig. S19.

Odor report data

Daily community odor report data were obtained upon request from the San Diego County Air Pollution Control District (SDAPCD). A simple linear regression analysis was conducted, yielding a Pearson correlation coefficient (*r*) of 0.92, indicating a strong positive correlation between H₂S levels and the number of daily odor reports.

HYSPLIT dispersion model

A series of forward trajectory simulations were run using the HYSPLIT dispersion model to study how local meteorology affects the observed H₂S levels as well as to estimate the relative levels of exposure in neighboring communities. HYSPLIT forward trajectories were started every hour at multiple heights within the estimated planetary boundary layer at 50-m intervals. The paths of all forward trajectories were then collected, weighted, and binned by time and location on the basis of assumed influences on emissions and levels. HYSPLIT trajectories were driven by the 12 km North American Mesoscale Forecast System (NAM).

To represent the influence of river flow and resulting turbulence on H₂S emissions, hourly Tijuana River flow rates at the start of each trajectory run were used as a weighting factor for eventual end point sums. Incorporation of flow was used to capture the effect of increased flow rates, which lead to higher source strengths and more turbulence

at the river hotspot, increasing H₂S water-to-air transfer. To account for photochemical loss of H₂S within the water, weights of trajectories initiated during daytime hours were downscaled assuming hourly losses at the water's surface following Pos *et al.* (76). To represent losses once in the atmosphere, trajectory weights were further scaled by length of daylight hours since release, assuming a H₂S daytime atmospheric lifetime of 12 hours and assuming chemical losses dominated by reaction with the hydroxyl radical OH. Each of these weighting factors represent highly simplified parameterizations of assumed influences on the emissions and losses of H₂S in this system, but they provide a quick test of those assumptions and their air quality implications. These assumptions qualitatively capture the measured variation in H₂S levels, further supporting the impact of river flow on H₂S levels.

Thermal desorption sampling and analysis

VOC sampling was conducted at three different sites in the Tijuana River Valley. Sampling details are outlined in table S1. Dual-bed [Tenax TA (35/60) and Carbograph 1 (60/80)] thermal desorption sorbent tubes (Camsco, Houston, TX, USA) were preconditioned at 335°C for 180 min while being backflushed with ultra-high-purity (UHP) nitrogen then stored at −15°C before deployment. VOCs were collected on the sorbent by sampling particle- and ozone-free ambient air at 200 standard cubic centimeters per minute (scm). Estuarine and coastal samples were collected for 60 min, whereas riverine samples were collected for 30 min. The difference in sample volumes was addressed in the data processing step by normalizing all samples to a default 60-min collection time. After collection, all sorbent tubes were temporarily (<15 min) placed in a portable cooler with blue ice packs and then stored at −15°C until laboratory analysis.

Samples were analyzed using two-dimensional gas chromatography time-of-flight mass spectrometry (GCxGC-ToF-MS, Pegasus 4D, LECO Corp., St. Joseph, MI, USA). Each sorbent tube was desorbed using an automated thermal desorber unit (ATD, TurboMatrix 650, PerkinElmer, Waltham, MA, USA), which introduced the volatilized sample onto the head of the primary GC column. The desorption procedure is detailed as follows. After being loaded into the ATD, each tube was leak tested and purged at ambient temperature for 5 min with UHP helium at 50 sccm to remove water and oxygen, reducing the risk of adsorbent oxidation during desorption. The purged contents flow through a cold (−20°C) focusing trap to ensure that no volatile analytes are lost during this step. Immediately after the tube purge, a gaseous internal standard mixture containing 1,2-dichlorobenzene-d₄ [1.441 parts per million by volume (ppmv)], 1-bromo-4-fluorobenzene (1.291 ppmv), fluorobenzene (2.255 ppmv), and toluene-d₈ (2.209 ppmv) in UHP nitrogen was automatically added to the tube. The tube was subsequently heated to 300°C, and the volatiles present in the sample were transferred at a helium flow rate of 50 sccm for 12 min to the cold focusing trap. After this step, the focusing trap was heated to 300°C for 5 min while being backflushed with helium, and 19% of the sampled mass was transferred at 1.2 sccm onto the head of the primary GC column. During this step, the primary GC column was held at 40°C for 6 min, allowing the sample to recondense for subsequent chromatographic separation. To ensure complete sample transfer from the ATD to the GCxGC-ToF-MS system, all valves and transfer lines upstream of the primary column were maintained between 225° and 235°C.

Chromatographic separation was obtained by ramping the primary GC column (DB-VRX 30 m by 0.25 mm by 1.4 μm, Restek Corporation, Bellefonte, PA, USA) from 40° to 220°C at a temperature rate of 4°C min^{−1}, while the effluents were collected, focused, and then transferred to the secondary GC column in 4-s modulation periods. The GC secondary column (Stabilwax 1.19 m by 0.25 mm by 0.5 μm, Restek Corporation, Bellefonte, PA, USA) and the modulator were offset by 5° and 20°C, respectively, in relation to the primary GC column. The ToF-MS detector was programmed to scan a mass/charge ratio range of 29 to 450, obtaining 200 spectra per second. The generated chromatograms were

processed using ChromaTOF v4.72.00 (LECO Corp., St Joseph, MI, USA). Compound identification was achieved by search with the National Institute of Standards and Technology (NIST) Mass Spectral Search Program. This study only included compounds whose mass spectra matched the database with a similarity score of 800 or higher. Kovats retention indices were also calculated (first dimension) and, when available, compared with those reported in the NIST database. We note that the available Kovats indices in the NIST database do not match the column specifications or mass spectrometry methods used in this work, and thus we report here the average of the available indices for nonpolar columns as a point of comparison. Single-ion integration was performed on compounds with a signal-to-noise ratio greater than 20. The raw abundances were corrected for instrument variability (i.e., detector drifts, sample transfer efficiency, etc.) using the internal standards response following the methodology described by Kreisberg *et al.* (77). Preconditioned nonmanipulated tubes were used for chromatogram blank subtraction to reduce the contribution of background noise to the integrated peak abundances. For each identified compound, available exposure guidelines were obtained from the Environmental Protection Agency (EPA) CompTox database. Acute Exposure Guideline Levels (AEGL) were prioritized, followed by Emergency Response Planning Guidelines (ERPG), and then by Temporary Emergency Exposure Limits (TEEL). Each guideline includes three levels of exposure values (e.g., AEGL-1, AEGL-2, and AEGL-3) for every chemical. Although there are important distinctions among the exposure guidelines, the tiers share some general similarities. The first tier represents a threshold for temporary, nondisabling effects. The second tier signifies a threshold for disabling effects (escape impairment). The third tier indicates a threshold for life-threatening effects. For each compound reported in this work, available concentrations for all three exposure guideline tiers can be found in tables S2 to S4.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Figs. S1 to S19; Tables S1 to S4; References (80, 81); Movie S1

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Heavily polluted Tijuana River drives regional air quality crisis

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Editor's summary

Intense industrial and urban wastewater pollution has plagued the Tijuana River for decades, causing long-term closure of neighboring beaches. Rico *et al.* showed that surging wastewater flows in the summer of 2024 enhanced water-to-air hydrogen sulfide (H₂S) transfer at a turbulent hotspot, leading to nighttime atmospheric H₂S peaks thousands of times greater than typical urban levels. This example makes it clear that poor water quality can drastically affect air quality—an observation with important implications for global waterways. —Jesse Smith

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